

# TANMOY SIL

## PERSONAL INFORMATION

**Tanmoy Sil**  
Rennweger Ring 15  
97070, Wuerzburg, Germany  
+49 152 220 66828  
tanmoy\_sil@hotmail.com  
LinkedIn · GitHub · Google Scholar

## SUMMARY

Biomedical engineer and machine learning researcher developing AI and sensing methods for movement disorders. Combines biosignal processing, deep learning, and computer vision with hands-on hardware and systems engineering, spanning DBS biomarker development, physiological sensing, and custom 3D motion capture systems, with experience translating methods from algorithm development to prospective clinical validation.

## EDUCATION

- |                  |                                                                                                                                                             |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2021–2026</b> | <b>PhD, Computational Neuroscience</b><br><i>University of Wuerzburg, Wuerzburg, Germany</i><br>Grade: 1.0 (magna cum laude)<br>PI: Prof. Dr. Jens Volkmann |
| <b>2018–2019</b> | <b>MS, Biomedical Engineering</b><br><i>University of Southern California, Los Angeles, USA</i><br>Grade: 3.52/4.0                                          |
| <b>2012–2016</b> | <b>BTech, Electronics &amp; Instrumentation Engineering</b><br><i>West Bengal University of Technology, Kolkata, India</i><br>Grade: 8.28/10.0              |

## RESEARCH EXPERIENCE

- |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                        |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>2026–</b>     | <b>Research Associate</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <i>University Clinic Wuerzburg, Wuerzburg, Germany</i> |
|                  | <ul style="list-style-type: none"><li>• Designed and built an 8-camera 3D motion capture laboratory for quantitative assessment of movement in stroke and spasticity research, integrating system architecture, optical design, hardware synchronization, real-time acquisition, and data infrastructure.</li><li>• Engineered an end-to-end overhead markerless 3D motion capture system for cervical dystonia research, including custom synchronization and acquisition hardware, real-time multi-camera recording, deep-learning-based pose estimation, multi-view triangulation, and synchronized video archival.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                        |
| <b>2021–2026</b> | <b>PhD Candidate</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <i>University Clinic Wuerzburg, Wuerzburg, Germany</i> |
|                  | <ul style="list-style-type: none"><li>• Developed a wavelet-based framework to detect and characterize beta bursts in Parkinson's disease STN local field potentials.</li><li>• Built a machine-learning classifier using EMG and accelerometry to distinguish Parkinson's disease from essential tremor, outperforming published baselines in sensitivity and specificity.</li><li>• Translated preclinical findings to human recordings by investigating relationships between beta-burst activity in the subthalamic nucleus and motor cortex, and the mediating role of striatal dopamine.</li><li>• Adapted a generative representation-learning framework to STN recordings during deep brain stimulation, disentangling stimulation-related activity from therapeutic response to identify a stimulation-orthogonal biomarker of motor improvement.</li><li>• Developed a video-derived Kinematic Discriminative Component biomarker that tracked medication and DBS effects more sensitively than MDS-UPDRS and was prospectively validated across two independent patient cohorts.</li></ul> |                                                        |

PUBLICATIONS

Koch, S.†, Perales, M.†, **Sil, T.†** et al. (2026). Probabilistic mouse–human brain correspondence by multimodal optimal transport [In Preparation].

Perales, M.†, **Sil, T.†**, et al. (2026). An interpretable latent motor axis tracks treatment and progression in Parkinson’s disease. *Nature Communications* [In Review].

**Sil, T.**, Navratil, P., et al. (2026). A stimulation-orthogonal neural signature of therapeutic response in Parkinsonian deep brain stimulation. *Science Advances* [In Submission].

Asadi, A., Peach, R., **Sil, T.**, et al. (2026). Parkinsonian beta bursts and deep brain stimulation: from subthalamic LFP electrophysiology to biophysical modelling. *npj Parkinson’s Disease* [In Review].

**Sil, T.**, et al. (2026). Sensor-based data-driven differentiation between Parkinson’s tremor and essential tremor. *Expert Systems with Applications*, 299, 130336. <https://doi.org/10.1016/j.eswa.2025.130336>

Hofman, K.†, Chen, J.Z.†, **Sil, T.**, et al. (2025). Low  $\beta$  predicts motor output and cell degeneration in the A53T Parkinson’s disease rat model. *Brain*, 2025: 00; 1–14. <https://doi.org/10.1093/brain/awaf063>

**Sil, T.**, et al. (2023). Wavelet-based bracketing, time-frequency beta-burst detection: new insights in Parkinson’s disease. *Neurotherapeutics*, 20, 1767–1778. <https://doi.org/10.1007/s13311-023-01447-4>

TEACHING & PROFESSIONAL EXPERIENCE

|           |                          |                                                                                                                                                                                                        |
|-----------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024–2025 | <b>Course Instructor</b> | <i>University Clinic Wuerzburg, Wuerzburg, Germany</i><br>Delivered lectures and hands-on coding tutorials for graduate-level data science courses during the summer semesters.                        |
| 2018–2019 | <b>Faculty Liaison</b>   | <i>USC Rossier School of Education, Los Angeles, USA</i><br>Managed admissions pipeline for 336 graduate students; streamlined faculty-administration coordination and improved processing efficiency. |
| 2018–2019 | <b>STEM Instructor</b>   | <i>USC Viterbi School of Engineering, Los Angeles, USA</i><br>Designed and delivered a hands-on engineering curriculum for 50 primary school students through USC Viterbi’s STEM outreach program.     |

TECHNICAL SKILLS

|                    |                                                                                                      |
|--------------------|------------------------------------------------------------------------------------------------------|
| <b>Research</b>    | Signal Processing, Machine Learning, Deep Learning, Computer Vision, Pose Estimation, Camera Systems |
| <b>Programming</b> | MATLAB, Python (PyTorch, scikit-learn, OpenCV, SciPy, NumPy)                                         |
| <b>Management</b>  | Project Management, Data Analysis, Cross-functional Collaboration                                    |
| <b>Languages</b>   | Bengali (native), English (fluent), Hindi (fluent), German (beginner)                                |

HONORS & AWARDS

|      |                                                                        |
|------|------------------------------------------------------------------------|
| 2022 | PhD Fellowship, Deutscher Akademischer Austauschdienst (DAAD), Germany |
| 2020 | Featured Graduate Student, USC Viterbi School of Engineering, USA      |
| 2015 | Best Intern Project Award, Variable Energy Cyclotron Centre, India     |

REFERENCES

Available upon request.